

Cpts 327 Assignment 6

Instructions

- **Format Requirement:** Please round (nor floor or ceil) all float answers to **two decimal places** (Please use fractions in intermediate steps to avoid precision loss and only round at the very end), e.g., round the result 4.555 to 4.56, 4.444 to 4.44. If your calculated answer has fewer than two decimal places (e.g., 0.1), you must pad it with a trailing zero (e.g., enter **0.10**) to ensure the grading system accepts it. **Please express all answers involving modular arithmetic (e.g., $(\text{mod } n)$) as the smallest integer in the range $[0, n-1]$.**
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Problem 1: (10 Points) Consider a textbook RSA cryptosystem where modulus $n := pq$ and $p = 11$ and $q = 13$.

- Calculate the Euler's totient function $\phi(n)$.
- If the public exponent is chosen as $e = 7$, calculate the private key d .
- Given a plaintext message $M = 5$, calculate the ciphertext C .
- Use d to decrypt a ciphertext $C' = 2$ to get the corresponding plaintext M' .

Problem 2: (10 Points) A commander wants to count his soldiers. When they stand in rows of 3, there are 2 left over. When they stand in rows of 5, there are 3 left over. When they stand in rows of 7, there are 2 left over. Let X be the number of soldiers.

$$X \equiv a_1 \pmod{3}$$

$$X \equiv a_2 \pmod{5}$$

$$X \equiv a_3 \pmod{7}$$

The CRT Formula: For $N = 3 \times 5 \times 7$, the solution is:

$$X = \sum_{i=1}^3 a_i \cdot M_i \cdot y_i \pmod{N}$$

Where a_i is the remainder, $M_1 = N/3$, $M_2 = N/5$, $M_3 = N/7$, and $y_1 = M_1^{-1} \pmod{3}$, $y_2 = M_2^{-1} \pmod{5}$, $y_3 = M_3^{-1} \pmod{7}$.

- Calculate y_1, y_2, y_3 .
- When $(a_1, a_2, a_3) = (2, 3, 2)$, calculate the smallest positive integer X .

Problem 3: (10 Points) RSA can be optimized using the CRT. Consider the RSA parameters $p = 17, q = 19, n = 323, e = 5$. Suppose we want to decrypt the ciphertext $C = 100$.

Perform the decryption $M = C^d \pmod{n}$ using the following CRT steps:

(a) Compute the private key d , and the reduced exponents:

$$d_p = d \pmod{p-1} \quad \text{and} \quad d_q = d \pmod{q-1}$$

(b) Compute the partial decryptions:

$$M_p = C^{d_p} \pmod{p} \quad \text{and} \quad M_q = C^{d_q} \pmod{q}$$

(c) Calculate the CRT coefficient $q_{inv} = q^{-1} \pmod{p}$.

(d) Recover the message $M = M_q + q \cdot [q_{inv}(M_p - M_q) \pmod{p}]$.

Problem 4: (10 Points) In a certain implementation of textbook RSA, the device computes $M_p = C^{d_p} \pmod{p}$ and $M_q = C^{d_q} \pmod{q}$ separately. An attacker interferes with the hardware such that the calculation of M_p has some mistakes, producing an incorrect value $\widetilde{M}_p$, while M_q remains correct. The device then uses the $\widetilde{M}_p$ and M_q to obtain a plaintext $\widetilde{M}$, following the procedure in Problem 3.

Given the following parameters:

- Modulus $n = 323$
- Public exponent $e = 5$
- Ciphertext $C = 100$
- "Plaintext" $\widetilde{M} = 158$

(a) Calculate $V = (\widetilde{M}^e - C) \pmod{n}$.

(b) Compute the greatest common divisor $g = \gcd(V, n)$.

(c) Calculate the two prime factors p and q .

(d) Calculate the real plaintext M .

Problem 5: (10 Points) True or False: Given a public key (n, e) whose n is sufficient large (e.g., 4096 bits), the textbook RSA encryption $C = M^e \pmod{n}$ is CPA-secure.

Problem 6: (10 Points) True or False: A variant of textbook RSA which appends a random 2-bits string r of to the message M before encryption, i.e., $C = (M||r)^e \pmod{n}$. This variant is CPA-secure.

Problem 7: (10 Points) True or False: Let $H : \{0, 1\}^* \rightarrow \mathbb{Z}_n^*$ be a secure hash function. A variant of textbook RSA whose encryption is defined as $C = H(M)^e \pmod{n}$ is CPA-secure.

Problem 8: (10 Points) True or False: Let a variant of textbook RSA encryption be $C = (r||M)^e \pmod{n}$, where r is a fresh random bit string chosen uniformly from $\{0, 1\}^k$ for each encryption. If k is sufficiently large against brute-force attack, the variant is CPA-secure.

Problem 9: (10 Points) Consider the following Search-LWE instance with $q = 100$, $n = 5$, and $m = 2$:

- Secret $\mathbf{s} = (s_1, s_2, s_3, s_4, s_5)^T \in \{0, 1\}^5$.
- Error $\mathbf{e} = (e_1, e_2)^T \in \{0, 1\}^2$.

The observed samples $(\mathbf{A}, \mathbf{b})$ are:

$$\begin{pmatrix} 48 & 24 & 12 & 6 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} \equiv \begin{pmatrix} 73 \\ 3 \end{pmatrix} \pmod{100}$$

Try to solve $\mathbf{s}$. You can do manual calculation using the hint or write some codes to solve it. (Hint: Multiple 1 is not a good sampling.)

Problem 10: (10 Points) Consider the following Search-LWE instance with $n = 3, m = 6$, and $q = 13$. The secret $\mathbf{s} = (s_1, s_2, s_3)^T \in \mathbb{Z}_{13}^3$ and the noise $e_i \in \{-1, 0, 1\} \pmod{13}$.

The elements in the observed matrix-vector samples $(\mathbf{A}, \mathbf{b})$ are:

$$\begin{pmatrix} 1 & 2 & 5 \\ 1 & 2 & 8 \\ 2 & 3 & 4 \\ 2 & 3 & 6 \\ 0 & 1 & 2 \\ 0 & 1 & 5 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \end{pmatrix} \equiv \begin{pmatrix} 1 \\ 4 \\ 6 \\ 8 \\ 8 \\ 11 \end{pmatrix} \pmod{100}$$

Try to solve $\mathbf{s}$. You can do manual calculation using the hint or write some codes to solve it. (Hint: Is it necessary to search the whole key space?)